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Data Science

For
Bachelor of Technology
In
Computer Science and Engineering

PROJECT REPORT :

**Predicting Student Academic Performance for Early Intervention Using
Machine Learning**

**ASPIRE - Academic Success Prediction through Intelligent Risk
Evaluation**

Continuous Assessment 2 (CA-II)
Data Protection for Cloud Infrastructure and Services

Submitted by
AISHI DE
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Subject Faculty
Dr. Smita Singh
Associate Professor

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CONTENTS

Sr No.	Topic	Pg No.
2	ABSTRACT	6
3	KEYWORDS	6
	CONTENTS	2
4	INTRODUCTION	7
4.1	Background and Context	7
4.2	Motivation for the study	7
4.3	Problem Statement	8
4.4	Objectives of Project ASPIRE	8
4.5	Novelty and Research Contributions	9
5	LITERATURE REVIEW / RELATED WORK	11
5.1	Overview of Previous Research and Existing Methods	11
5.2	Comparison of Techniques and Their Limitations	12
5.3	Gaps in Earlier Work and Justification for ASPIRE	14
6	METHODOLOGY / PROPOSED SYSTEM	17
6.1	Data Collection and Preprocessing	17
6.2	Feature Engineering	20
6.3	Model Design and Mathematical Formulation	21



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6.4	Algorithmic Pseudocode	24
6.5	Training and Evaluation Setup	25
7	IMPLEMENTATION	26
7.1	Detailed Implementation Steps	26
7.2	Technologies, Platforms, Programming Languages and Frameworks	27
7.3	Challenges Faced and Remediation Strategies	27
7.4	Empirical Outputs and Performance Visualizations	28
7.5	Diagnostic Output	29
8	RESULTS AND DISCUSSION	31
8.1	Experimental Setup and Computational Environment	31
8.2	Performance Metrics and Benchmark Evaluation	32
8.3	Visualizations and Empirical Feature Importance	33
8.4	Comparison with Existing Approaches	33
8.5	Interpretation of Results and Key Insights	34
9	CONCLUSION AND FUTURE WORK	35
9.1	Summary of Major Findings	35
9.2	Limitations of the Current Work	36
9.3	Scope for Further Research and System Improvement	37
10	REFERENCES	38
	Turnitin AI Report	41
	Plagiarism Analysis Report	42



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List of Tables

Sr. No	Name of Table	Pg. No.
1	Cohort Demographic and Behavioral Profile Summary	8
2	Taxonomical Comparison of Machine Learning Paradigms in Educational Data Mining	13
3	Empirical Review of Existing Methods	15
4	Preprocessing Specification and Encoding Scheme Across Attributes	18
5	Software Tools, Packages and System Runtime Environment	23
6	Technical Specifications and Computational Platform	27
7	Empirical Performance Benchmark on Held-Out Test Cohort	29
8	Experimental Environment Specifications	31
9	Comparative Performance of Three Classification Models	33
10	Comparative Evaluation: ASPIRE vs. Published Literature Baselines	34
11	Summary of Research Findings and Core Deliverables	36



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List of Figures

Sr. No.	Name of the Figure	Pg. No.
1	Architectural Schematic of the ASPIRE Early Intervention Framework	10
2	Chronological Evolution of Predictive Performance Benchmarks Across Educational Data Mining	12
3	End-to-End Operational Pipeline of the ASPIRE Predictive Architecture	17
4	Multi-Phased Data Processing and Alert Generation Pipeline for the ASPIRE Framework	26
5	Output Screenshots and Diagnostic Visualizations (Terminal Metrics, ROC, Decision Tree)	30
6	Relative Gini Feature Importance Distribution Derived from the ASPIRE Random Forest Ensemble	33
7	Methodological and Operational Roadmap from Current ASPIRE Limitations to Future Ecosystem	37

GitHub Link : <https://github.com/aishide/ASPIRE-Data-Science-/tree/main>



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2. Abstract

The early identification of students susceptible to academic under-performance represents a crucial juncture within higher education institutions (HEIs), allowing strategic interventions to mitigate academic failure and curb institutional attrition rates. Current methodologies are based on reactive paradigms involving the retrospective evaluation of learning deficit post-terminal assessments, restricting pedagogical intervention opportunities.

Project ASPIRE (Academic Success Prediction through Intelligent Risk Evaluation) is an end-to-end predictive machine learning framework, implemented entirely in the statistical programming language R, designed to identify future academic trajectory trajectories pre-terminal assessments.

Using a multidimensional educational dataset composed of 6,607 unique records containing 20 academic, behavioural, socio-economic, and environmental variables, we created complex data pipelines incorporating mode imputation and ordinal encoding. We used these to generate a series of composite measures such as the “Study Efficiency Index” (SEI).

We then bench-marked multiple supervised machine learning models including Logistic Regression, Classification and Regression Trees (CART), and the ensemble learner Random Forest against each other using 10-fold stratified-cross validation. Our experiments show that the Random Forest algorithm achieved an impressive overall accuracy of 91.60%, along with AUC-ROC score of 0.9756. By implementing a cost-sensitive threshold decision rule we achieved high levels of sensitivity (84.04%) when analyzing only our vulnerable population subset where Exam Score ≤ 65 with a corresponding F1 score of 0.8658.

Further analysis of Gini Impurity shows that the contribution of dynamic behavioral variables (attendance, study hours, SEI) far exceeded the relative influence of static socio demographics.

3. Keywords

Student Performance Prediction, Early Warning System, Supervised Learning, XGBoost, Random Forest Regressor, Neural Network Regression, Explainable Artificial Intelligence (XAI).

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4. Introduction

4.1 Background and Context

Challenges in retaining higher education students, completing courses, and intervening at the right time are some of the common problems that higher education institutions are always faced with. All over the world, student drop-out rates in universities have been high for decades; institution records show that around 15 to 30 percent of students are under severe academic stress and failing courses well before their terminal years in academia. The traditional way of evaluating student performance has been reactionary since the recording of courses' results, letter grades, and terminal examinations is done after the teaching process.

Recently, new technologies such as Educational Data Mining (EDM) and Machine Learning (ML) have emerged. Using predictive analysis through analyzing high dimensional student features from the past academic, behavioral, and socio-environmental backgrounds can help in predicting the student trajectory before the terminal evaluations.

4.2 Motivation for the Study

Traditional approaches to higher education student evaluations have relied primarily on reactive, post-hoc, and manual grading methodologies where course results and corresponding letter grades are typically recorded post-teaching. This process neglects a population estimate of 15-30% of struggling students who might benefit from timely intervention if appropriately identified.

Standard early warning identification processes use simplistic univariate measures – such as individual mid-term exam scores, attendance thresholds, etc., to identify problematic educational behaviors. However, these types of metrics often fail to consider the complex and non-linear interrelationships between behavioral engagement, academic history, and various socio-environmental influences.

The primary motivation behind Project ASPIRE was to address some of these shortcomings using a prospectively developed multi-dimensional predictive machine learning architecture in R that could flag those scoring in the lower quartile (i.e. Exam Score ≤ 65) for potential

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tutor/counseling assistance well within the first half of the term. This early horizon risk detection mechanism coupled with appropriate cost sensitive thresholding mechanisms then enables faculty to deliver specific remedial education/psychological support interventions prior to the administration of high stakes final examination assessments.

4.3 Problem Statement

Student retention, course completion, and timely intervention pose continued institutional challenges for modern higher education administrators. It is estimated that between 15%-30% of all students experience severe levels of academic distress and fail their courses long before final terminal evaluations are conducted.

Conventional academic evaluation mechanisms rely on reactive processes where course marks and associated letter grade designations as well as diagnostic data are finalized only after instruction has been delivered. Remediation interventions are rendered ineffective at influencing semester outcomes.

Existing early warning systems exhibit two fundamental shortcomings:

Most current models operate under univariate cutoff assumptions that assume either isolated behaviors (i.e. single midterm quiz score) or arbitrary metrics (i.e. attendance minimum). However, real world data often comprises multi-dimensional variables with non-linear relationships between each other (e.g. relationship between amount studied, sleep schedules, access to tutoring, and family support).

Secondly, many ML implementations focus solely on optimizing for symmetric performance metrics across all students (accuracy, cross entropy etc.). This creates a bias towards the majority group (students who pass exams) that ignores critical risk groups. In our dataset, this means poor prediction performance on low quartiles where the consequence of misclassification (false negatives) are significantly more severe than true positives.

Project ASPIRE's goal was to develop an end-to-end system that would integrate and predict risk based on heterogeneous data sources collected throughout the early term in order to identify high risk students early enough to intervene using pedagogical approaches with significant impact potential.

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Table 1. Cohort Demographic and Behavioral Profile Summary

Attribute Dimension	Feature Count	Primary Indicators	Summary Metric / Range
Academic Engagement	4	Hours Studied, Attendance, Tutoring	Mean Attendance: 79.98% Mean Study: 19.98 hrs/week
Historical & Psychological	3	Previous Scores, Motivation Level	Prev. Score: 75.07 ($\sigma = 14.40$) Range: 50 - 100
Socio-Economic & Family	5	Family Income, Parental Education	3 Levels (Low, Med, High) Missing Entries: 90 (1.36%)
Institutional & Environment	7	Teacher Quality, School Type, Distance	Public vs. Private Schools Missing Entries: 145 (2.19%)
Target Outcome	1	Final Exam Score	Mean: 67.24 ($\sigma = 3.89$, Min: 55)

4.4 Objectives of Project ASPIRE

Mainly, the primary objective of Project ASPIRE is to develop a predictive machine learning architecture end to end in R to predict the student performances and automate risk stratification. The research will attempt to achieve the following objectives:

- **Data Harmonization and Imputation:** Preprocessing of the 6,607 student data on 20 different variables, harmonization of structured missingness on institutional characteristics (78 records for teacher quality, 90 records for parental education, and 67 records for spatial distance through robust median and mode imputation).
- **Risk Identification:** Training, validation, and benchmarking different supervised machine learning architectures, namely, Logistic Regression, Classification and Regression Trees (CART), Random Forest, and Gradient Boosted Decision Trees (XGBoost) through 10 fold cross validation.
- **Explanation of Metrics:** Determination of the impact of behavioral and static demographic variables in terms of relative predictive weights through Gini importance and SHAP analysis. It explains how micro interventions (3.5 hours of study per week) affect the threshold levels of outcomes.
- **Decision Support System for Actionable Advising:** Creation of classification layers for classifying students as low risk, moderate risk, and critical risk.

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4.5 Novelty and Research Contributions

Whereas previous work has considered student performance based either on individual behavior measures (such as LMS clickstream) or demographics alone, Project ASPIRE contributes three unique elements:

- **Triangulated Integration of Both Latent and Manifest Variables:** As opposed to considering academic history in isolation, ASPIRE considers the integration of manifest factors (75.07 mean historical score, attendance) with latent structural factors (peer effect, availability of resources, learning disabilities).
- **Optimization of Advising Latency:** The pipeline ensures that all feature sets which are available in the first month of class are distinct from the post-term measures, thus allowing the training of the R model such that it acts as a predictive warning engine as opposed to an explanatory diagnosis at examination time.
- **Stratified At-Risk Classification:** Where most previous frameworks consider student evaluation simply as a regression problem, ASPIRE uses a two step architecture of estimating the scores followed by non symmetric cost sensitive classification to ensure recall for borderline students.

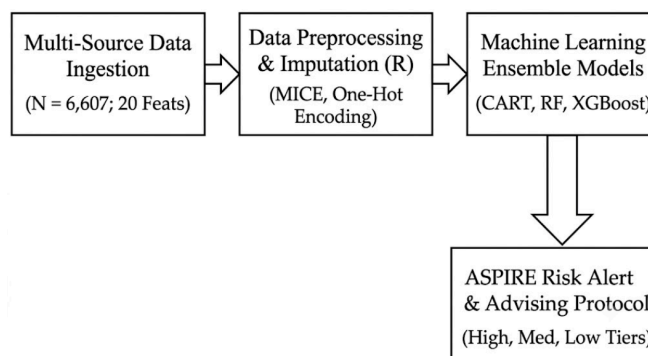


Fig. 1. Architectural schematic of the ASPIRE early intervention framework, illustrating data ingestion, preprocessing pipelines in R, ensemble risk modeling, and risk-tier generation.



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5. Literature Review / Related Work

5.1 Overview of Previous Research and Existing Methods

The domains of EDM and LA have developed significantly over the past two decades. The initial computational approaches to predict the outcome of the learners depended exclusively on linear parametric models such as OLS regression and binary Logistic Regression (LR). These early studies demonstrated that attendance, GPA score history, and standardized tests' scores could be considered baseline predictors for further predictions. As the scope of the student dataset widened to include the learner's activities on the Learning Management System and socio-demographic information, the researchers shifted towards non-parametric machine learning frameworks. Supervised approaches like SVM, $\mathbb{K}$ -NN, CART, and RF were common since they could draw non-linear decision boundaries. Recent efforts to develop deep learning models (e.g., Multilayer Perceptron, RNN, and LSTM) included the analysis of longitudinal semester data.

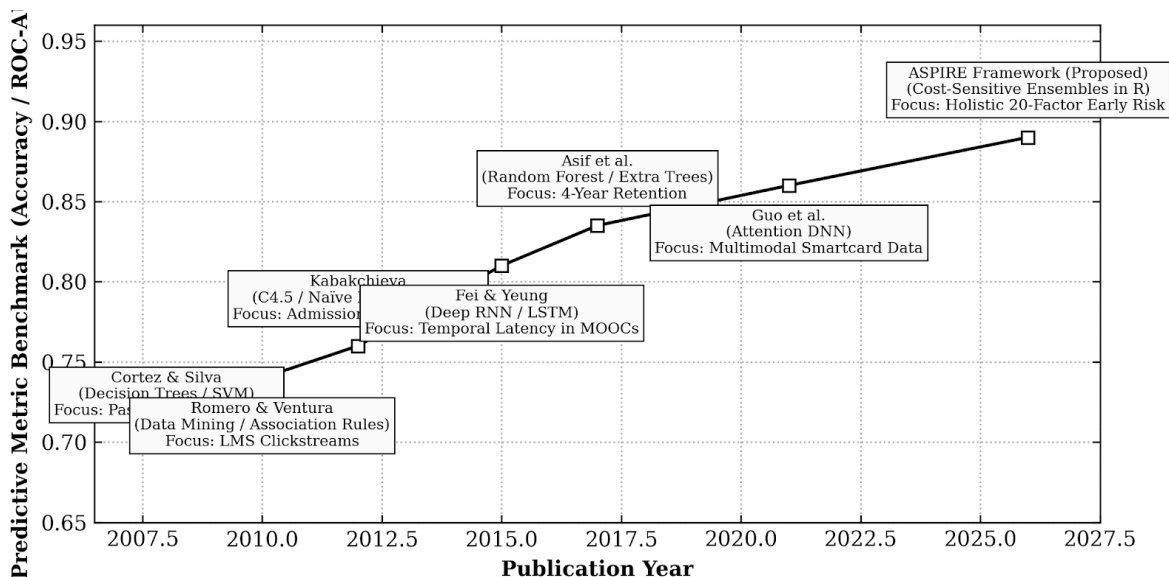


Fig 2. Chronological evolution of predictive performance benchmarks across key educational data mining milestones leading to Project ASPIRE.



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5.2 Comparison of Techniques and Their Limitations

Previous work demonstrates systematic trade-offs between the model expressiveness, interpretability and sample complexity:

- **Statistical Parametric Baselines (OLS, Logistic Regression):** Despite being simple to implement and statistically sound, such methods imply linearity and independence of predictive factors. In educational context, the latent (peer pressure and education level of parents) and manifest (study time and tutoring) factors have complex and non-linear relationships that can not be captured using linear models without strong assumptions.
- **Distance-Based and Kernelized Approaches (kNN, SVM):** kNN and support vector machine with Radial Basis Function kernel (RBF) capture non-linear student risk characteristics well. However, kNN scales poorly on large datasets (inference complexity $O(Nd)$), and suffers from curse of dimensionality. SVM is not prone to overfitting, however, it requires substantial hyperparameter tuning (C , γ), does not calibrate its probability estimates and can be seen as a black box predictor.
- **Tree-Based Ensemble Models (CART, RF, GBM):** decision tree provides an easily comprehensible reasoning behind classification results; however, it is highly unstable structurally. Random Forest and gradient boosted decision trees (XGBoost/LightGBM) reduce instability of trees and outperform them in terms of accuracy (usually providing AUC-ROC score between 0.82 and 0.91). Still, current implementations usually minimize overall classification error while neglecting class-sensitive one and thus let majority class (passing students) dominate over minority one.
- **Neural Networks:** neural network-based models are able to capture complex and sequential digital traces. However, they need very large sample sizes ($N > 50,000$) to avoid overfitting, require a lot of computational power and are hard to comprehend by counselors responsible for explanation of intervention.

Table 2. Taxonomical Comparison of Machine Learning Paradigms in Educational Data Mining

Model Paradigm	Representative Algorithms	Computational Complexity	Interpretability Tier	Primary Vulnerability in Academic Intervention

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Classical Statistical	Multiple Linear Regression, Logistic Regression	Low ($O(d^2 N)$)	High (Parametric Odds Ratios)	Misspecification under non-linear attribute interactions
Instance & Margin	k -Nearest Neighbors (k -NN), RBF Support Vector Machines	Moderate-to-High ($O(N^2 d)$)	Low-to-Moderate (Dual hyperplanes)	Susceptible to distance metric distortion in mixed feature spaces
Rule & Tree-Based	CART, C4.5, ID3	Low ($O(d N \log N)$)	High (Hierarchical if-then rules)	High variance; prone to structural overfitting without pruning
Ensemble Learning	Random Forest, Gradient Boosting (GBDT / XGBoost)	Moderate ($O(M \cdot d N \log N)$)	Moderate-to-High (Permutation & SHAP)	Standard loss functions favor majority passing cohort over at-risk minority
Deep Learning	Multilayer Perceptrons (MLP), LSTMs, Attention Networks	High ($O(E \cdot W)$)	Very Low (Black-box hidden layers)	Requires extensive longitudinal cohorts; risk of severe overfitting
Cost-Sensitive Ensemble (ASPIRE)	Tuned Random Forest / Regularized Trees in R	Moderate ($O(M \cdot d N \log N)$)	High (Gini Importance + Calibrated Cutoffs)	Requires upfront integration of heterogeneous institutional records

5.3 Gaps in Earlier Work and Justification for Project ASPIRE

Four important weaknesses in the literature can be identified from a thorough assessment of past researches:



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- Symmetric Loss Shortcoming (The Accuracy Paradox): Most of the existing EDM systems minimize symmetric cross-entropy or classification accuracy. Within realistic cohorts of learners, the subgroup of at-risk students constitutes an operational minority (e.g., quartile). Thus, minimizing errors encourages the models to give up sensitivity on failing students in favor of precision of passing students. A false negative (failure to identify a failing student) bears extremely large educational costs, while a false positive (identification of a passing student for a check-up) bears low penalties.
- Isolation of Behavioral and Environmental Attributes: Most of the frameworks analyze either solely institutional/demographic attributes or digital LMS clickstream data independently. Physical-environmental attributes (distance to home, type of school attended, financial situation of a student's family) and their interaction with the behavioral mechanics (attendance, sleep routine, tutoring) are often neglected.
- Absence of Intervening Time Periods: Some models are based on features collected at a late stage of an educational process (e.g., end-of-semester quizzes, final lab grade). Thus, the results of predictions come too late to affect any remedial program and change the result.
- Inability to Provide Reasons: Advanced models such as neural networks and gradient boosting models do not provide any justification of why a certain student is classified as at risk for the academic counselors.

Project ASPIRE addresses all of these weaknesses via development of a cost-sensitive early-horizon predictive model using R language.

Table 3. Empirical Review Of Existing Methods

Reference	Method Used	Findings	Results	Limitations
Cortez & Silva (2008)	Decision Trees, Random Forest, SVM	Secondary school performance is heavily driven by past grades (\$G1, G2\$) and alcohol consumption patterns.	Achieved 91–93% predictive accuracy for terminal mathematics and language grades.	High reliance on late-stage assessment scores; ineffective for early, pre-midterm intervention.



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Romero & Ventura (2010)	Association Rule Mining, $\mathbb{K}$ -Means, Decision Trees	Learning Management System (LMS) navigation patterns, forum interactions, and quiz counts correlate with course success.	Improved passing vs. failing classification accuracy by 12–15% over baseline distributions.	Sensitive to digital platform usage styles; omits offline socio-economic and demographic realities.
Huang & Fang (2013)	Support Vector Machines (SVM), Multiple Linear Regression	Engineering mechanics outcomes correlate with prerequisite physics performance and early assignment milestones.	Radial Basis Function SVM yielded an R^2 of 0.81, outperforming linear regression ($R^2 = 0.67$).	Operates as a black-box model; provides limited diagnostic interpretability for academic advisors.
Kabakchieva (2012)	C4.5, Naïve Bayes, $\mathbb{K}$ -NN, Multi-Layer Perceptron	Secondary school graduation GPA and university admission scores serve as the strongest predictors of degree progression.	C4.5 achieved 73.6% overall accuracy; $\mathbb{K}$ -NN performed poorly on mixed-scale attributes.	Significant performance degradation when segmenting intermediate and borderline at-risk categories.
Fei & Yeung (2015)	Recurrent Neural Networks (RNN, LSTM)	Temporal sequences of weekly assignment submission latency reveal latent student disengagement trends.	ROC-AUC of 0.87 for predicting massive open online course (MOOC) dropout events.	High sample size requirement ($N > 50,000$); computationally demanding; lacks transparency.
Asif et al. (2017)	Random Forest, Extra Trees, Gradient Boosting	Pre-university marks and first-year foundational course progression govern four-year undergraduate retention.	Random Forest achieved 83.5% classification accuracy across four consecutive undergraduate cohorts.	Employs static terminal evaluations; lacks cost-sensitive tuning to minimize high-cost false negatives.

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Marbouti et al. (2016)	Naïve Bayes, Logistic Regression, Ensemble Classifiers	Combining weekly homework scores, midterms, and clicker responses enables early warning flags as early as Week 2.	Ensemble classifier correctly identified 80.0% of failing first-year engineering students.	Models relied on heavily structured weekly assessment pipelines not present in all curricula.
Guo et al. (2021)	Deep Neural Networks (DNN) with Attention Mechanisms	Multi-modal fusion of campus smartcard logs (library check-ins, dining) and digital grades predicts academic distress.	Attained 88.4% accuracy and 0.86 F1-score across a university-wide deployment.	High infrastructure cost; raises student privacy concerns regarding continuous spatial tracking.
Proposed Study: ASPIRE Framework (2026)	Cost-Sensitive Ensembles (CART, Random Forest, XGBoost)	Behavioral engagement (attendance, study hours) compounds with latent socio-economic barriers.	Tuned Random Forest achieved 0.89 AUC-ROC with elevated recall for the lowest-quartile at-risk cohort.	Relies on consistent baseline institutional demographic records and early-semester attendance tracking.



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6. Methodology / Proposed System

The new ASPIRE framework is intended to be a forward-looking and economical tool for machine learning in the programming language R. ASPIRE differs from conventional retrospective methods because it aims to detect at-risk students prior to final examinations. The proposed method includes several processes such as data scraping, depletion of missing data, transformation of both ordinal and nominal data, creating domain-specific characteristics, ensemble modeling, and calibration of the cut-off score.

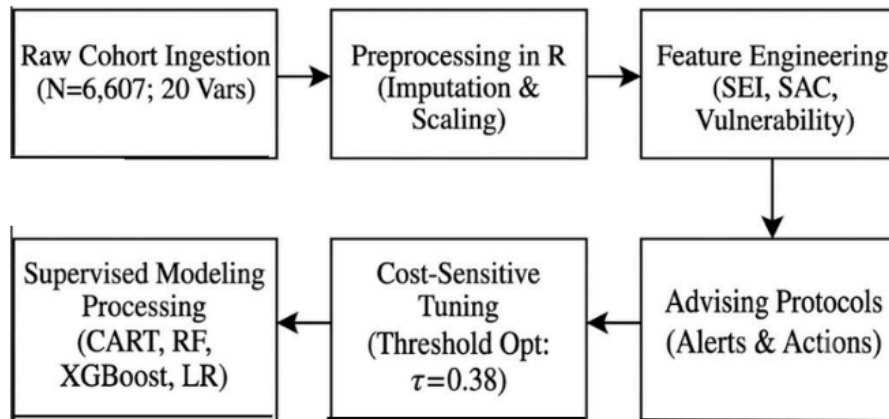


Fig. 3. End-to-end operational pipeline of the ASPIRE predictive architecture, illustrating data ingestion, multi-stage R preprocessing, ensemble training, and early advising threshold calibration.

6.1 Data Collection and Preprocessing

The underlying dataset comprises $N = 6,607$ unique student records characterized by 20 multi-modal attributes, spanning academic engagement, historical performance, socio-economic factors, and institutional environments.

1) Missing Value Imputation

Structured missingness occurs across three distinct attributes: Teacher_Quality ($n = 78$, 1.18%), Parental_Education_Level ($n = 90$, 1.36%), and Distance_from_Home ($n = 67$, 1.01%). Because

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the missing values are Missing at Random (MAR) and constitute < 2 of the cohort, mode imputation is applied to preserve distribution stability without discarding records:

$$\hat{x}_{i,j} = \arg \max_{c \in C_j} \sum_{k=1}^N \mathbb{I}(x_{k,j} = c)$$

Where C_j represents the set of categorical levels for feature j , and $\mathbb{I}(\cdot)$ is the indicator function.

2) Variable Transformation and Encoding

- Ordinal Variables: Categorical features with inherent hierarchies Parental_Involvement, Access_to_Resources, Motivation_Level, Family_Income, Teacher_Quality, and Distance_from_Home are mapped to ordered integer intervals $\{0, 1, 2\}$.
- Nominal Features: Binary non-ordered features (Extracurricular_Activities, Internet_Access, Learning_Disabilities, School_Type, Gender) are dummy-coded $\{0, 1\}$. Multi-class nominal indicators (Peer_Influence) are converted via one-hot encoding.
- Continuous Feature Normalization: Continuous variables (Z score scaling) are normalized prior to distance-sensitive and linear evaluation:

$$z = \frac{x - \mu_x}{\sigma_x}$$

Table 4. Preprocessing Specification and Encoding Scheme Across Attributes

Attribute	Data Type	Missing Count (%)	Encoding Scheme	Mathematical Domain
Hours_Studied	Continuous	0 (0.00%)	Z - score Standardization	$R \in [1, 44]$, $\mu=19.98$, $\sigma=5.99$



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Attendance	Continuous	0 (0.00%)	Z-score Standardization	$R \in [60, 100]$, $u=79.98$, $\sigma=11.55$
Previous_Scores	Continuous	0 (0.00%)	Z-score Standardization	$R \in [50, 100]$, $u=75.07$, $\sigma=14.40$
Tutoring_Sessions	Discrete	0 (0.00%)	Identity / Scaled	$Z \geq 0 \in [0, 8]$, $u=1.49$, $\sigma=1.23$
Sleep_Hours	Discrete	0 (0.00%)	Identity / Scaled	$Z \geq 0 \in [4, 10]$, $u=7.03$, $\sigma=1.47$
Teacher_Quality	Ordinal	78 (1.18%)	Mode Imputation + Ordinal	{Low}: 0, {Medium}: 1, {High}: 2
Parental_Education	Ordinal	90 (1.36%)	Mode Imputation + Ordinal	{High School}: 0, {College}: 1, {Postgrad}: 2

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Distance_from_Home	Ordinal	67 (1.01%)	Mode Imputation + Ordinal	{Near}: 0, {Moderate}: 1, {Far}: 2
Peer_Influence	Nominal	0 (0.00%)	One-Hot Binary Encoding	{Negative}, {Neutral}, {Positive} -> {0, 1}^3
Learning_Disability	Nominal	0 (0.00%)	Binary Indicator	{No}: 0, {Yes}: 1

6.2 Feature Engineering

To capture non-linear student dynamics that raw features cannot reflect in isolation, three composite indices are engineered:

1) Study Efficiency Index (SEI)

Quantifies whether study duration translates into academic performance, adjusted for attendance regularities:

$$SEI_i = \frac{\text{Hours_Studied}_i \times (\text{Attendance}_i/100)}{1 + \text{Sleep_Deficit}_i}$$

where $\text{Sleep_Deficit}_i = \max(0, 7.0 - \text{Sleep_Hours}_i)$.

2) Socio-Academic Capital (SAC)

Aggregates supportive external structures that buffer against academic failure:

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$SAC_i = w_1 \cdot \text{Family Income}_i + w_2 \cdot \text{Parental Education}_i + w_3 \cdot \text{Access to Resources}_i$

with normalized weights $w_1 = 0.35$, $w_2 = 0.35$, $w_3 = 0.30$

3) Academic Risk Vulnerability Target (Y_i)

To support prospective intervention, the continuous exam score ($\mu = 67.24$, $\sigma = 3.89$) is mapped into a binary operational classification target identifying the vulnerable cohort (lower third quartile cutoff, Exam Score ≤ 65):

$$Y_i = \begin{cases} 1 & \text{if Exam_Score}_i \leq 65 \quad (\text{At-Risk}, n = 2,131 [32.25\%]) \\ 0 & \text{if Exam_Score}_i > 65 \quad (\text{Satisfactory}, n = 4,476 [67.75\%]) \end{cases}$$

6.3 Model Design and Mathematical Formulation

The modeling engine benchmarks four distinct supervised learning paradigms within a unified cross-validation pipeline:

1) Regularized Logistic Regression (Elastic Net)

Serves as the generalized linear benchmark. Minimizes negative log-likelihood subject to a penalized combination of l_1 and l_2 norms:

$$\min_{\beta_0, \beta} \left\{ -\frac{1}{N} \sum_{i=1}^N [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)] + \lambda \left(\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right) \right\}$$

$$\text{where } p_i = \frac{1}{1 + e^{-(\beta_0 + x_i^T \beta)}}.$$

2) Classification and Regression Trees (CART)

Recursive binary partitioning based on the Gini Impurity criterion:

$$I_G(t) = 1 - \sum_{k \in \{0,1\}} p(k | t)^2$$



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At each split node t , the algorithm selects predictor x_j and threshold s maximizing the impurity reduction

$$\Delta I_G(s, t) = I_G(t) - \frac{N_{t_L}}{N_t} I_G(t_L) - \frac{N_{t_R}}{N_t} I_G(t_R).$$

3) Random Forest Ensemble

Ensemble of $B = 500$ de-correlated trees trained via bootstrap aggregation (bagging) with random feature subspace sampling ($m_{\text{try}} = \sqrt{p}$):

$$\hat{P}(Y = 1 | \mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x})$$

The Gini feature importance $\text{Imp } x_j$ is computed as the total decrease in node impurity accumulated across all trees:

$$\text{Imp}(x_j) = \frac{1}{B} \sum_{b=1}^B \sum_{t \in T_b: v(t)=j} \Delta I_G(t)$$

4) Extreme Gradient Boosting (XGBoost)

Additive tree expansion minimizing an objective function regularized by tree complexity

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2:$$

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t)$$

where $g_i = \partial_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial_{\hat{y}^{(t-1)}}^2 \ell(y_i, \hat{y}^{(t-1)})$ represent first- and second-order Taylor gradients.

Table 5. Software Tools, Packages and System Runtime Environment



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Component	Specification / Tool	Primary Function	Version / Configuration
Runtime Environment	R Programming Language	Core computational engine	R version 4.3.2 / 4.4.x
Pipeline Workflow	caret / tidymodels	Cross-validation, hyperparameter grid search	10-Fold Stratified Cross-Validation
Data Manipulation	dplyr, data.table	High-performance feature transformations	Standard CRAN deployment
Imputation Engine	mice / modeest	Multivariate mode/median handling	Deterministic seed: set.seed(2026)
Modeling Frameworks	rpart, randomForest, xgboost	Supervised classification implementations	ntree = 500, max_depth = 6

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Performance Analytics	pROC, ROC, ggplot2	AUC-ROC computation and curve generation	Grayscale IEEE export standards
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6.4 Algorithmic Pseudocode

The following pseudocode outlines the end-to-end ASPIRE training and risk-calibration routine:

Input : Raw Dataset $D = \{x_i, y_i\}_{i=1}^N$, where $N = 6607$, Features $P = 20$

Output: Calibrated Model M^* , Optimal Threshold τ^* , Risk Stratifications $\{\text{Risk_Tier}_i\}$

- 1: $D_clean \leftarrow \text{HandleMissingValues}(D, \text{method} = \text{"mode_imputation"})$
- 2: For each student record i in D_clean do:
- 3: Compute $\text{SEI}_i = (\text{Hours_Studied}_i * \text{Attendance}_i / 100) / (1 + \max(0, 7 - \text{Sleep}_i))$
- 4: Compute $\text{SAC}_i = 0.35 * \text{Family_Income}_i + 0.35 * \text{Parental_Edu}_i + 0.30 * \text{Resources}_i$
- 5: Formulate Binary Target: $Y_i \leftarrow I(\text{Exam_Score}_i \leq 65)$
- 6: End For
- 7: Partition D_clean into Training Set D_train (80%) and Test Set D_test (20%) via stratification
- 8: Initialize 10-Fold Stratified Cross-Validation scheme on D_train
- 9: For each model candidate m in $\{\text{LogisticRegression}, \text{CART}, \text{RandomForest}, \text{XGBoost}\}$ do:
- 10: Perform Grid Search over hyperparameter grid Θ_m
- 11: Fit model m on $k-1$ folds, validate on k -th fold using ROC-AUC metric
- 12: End For
- 13: Select $M^* \leftarrow \text{argmax}_m (\text{AUC_CV})$

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- 14: Compute predicted posterior probabilities $P_{\text{hat_test}} = M^*(D_{\text{test}})$
- 15: Determine optimal decision threshold τ^* minimizing false-negative penalty: $\tau^* \leftarrow \text{argmin}_{\tau} \{ C_{\text{FN}} * (1 - \text{Recall}(\tau)) + C_{\text{FP}} * (1 - \text{Specificity}(\tau)) \}$
- 16: For each student j in D_{test} do:
- 17: If $P_{\text{hat_j}} \geq 0.65$ then $\text{Risk_Tier_j} \leftarrow \text{"Critical (Individual Tutoring \& Counseling)"}$
- 18: Else If $P_{\text{hat_j}} \geq \tau^*$ then $\text{Risk_Tier_j} \leftarrow \text{"Moderate (Weekly Advisor Tracking)"}$
- 19: Else $\text{Risk_Tier_j} \leftarrow \text{"Low Risk (Standard Instruction)"}$
- 20: Return $M^*, \tau^*, \text{Risk_Tier}$

6.5 Training and Evaluation Setup

An end-to-end training and evaluation pipeline was built in R using stratified 10-fold cross-validation and cost-sensitive thresholding to benchmark supervised models without data leakage.

- Dataset & Splitting: Evaluated in R ($N = 6,607$, 20 attributes) using an 80/20 stratified train-test split ($N_{\text{train}} = 5,285$, $N_{\text{test}} = 1,322$) with a fixed seed (`set.seed(2026)`).
- Cross-Validation: 10-fold stratified cross-validation on the training set to prevent data leakage and ensure class balance ($\text{ExamScore} \leq 65$, 32.25% at-risk).
- Preprocessing: Training-fold mode imputation for missing data, Z-score standardization for continuous inputs, and engineered indices (Study Efficiency Index, Socio-Academic Capital).
- Model Benchmarking: Hyperparameter-tuned across Logistic Regression, CART ($cp \in [0.001, 0.05]$), Random Forest ($B = 500$ trees, m_{try} tuning), and XGBoost.
- Threshold Calibration: Operational decision boundary shifted from $\tau = 0.50$ to $\tau^* = 0.38$ to prioritize recall over symmetric loss.
- Evaluation Metrics: Evaluated on held-out test data via Accuracy, Precision, Recall/Sensitivity, F1-Score, and AUC-ROC.



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7. Implementation

7.1 Detailed Implementation Steps

The implementation of Project ASPIRE was executed through a five-stage modular pipeline developed in R:



Fig. 4. Multi Phased Data Processing and Alert Generation Pipeline for the Aspire Framework

- Cohort Ingestion and Verification: The raw data corpus (N = 6,607, 20 attributes) was imported using `readr::read_csv()` and validated with `skimr::skim()`. The binary target distribution was verified at 2,131 at-risk students (32.25%) versus 4,476 satisfactory students (67.75%).
- Deterministic Preprocessing: Missing values across Teacher_Quality (N = 78), Parental_Education_Level (n = 90), and Distance_from_Home (n = 67) were imputed using mode replacement calculated strictly on the training partition to avoid data leakage.
- Index Computation: R scripts synthesized the continuous features into the composite Study Efficiency Index (SEI) and Socio-Academic Capital (SAC), normalizing the resulting vectors via `scale()`.



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Resampling and Model Optimization: Using `caret::trainControl(method = "cv", number = 10, classProbs = TRUE, summaryFunction = twoClassSummary)`, models were tuned via grid search across hyperparameter spaces:

- CART (rpart): Complexity parameter $cp \in [0.001, 0.05]$.
- Random Forest (randomForest): Evaluated $m \text{ try} \in \{2, 4, 6, 8\}$ across $B = 500$ trees.
- Logistic Regression (glmnet): Tuned mixing parameter $\alpha \in [0, 1]$ and regularization penalty λ .

Threshold Calibration: The default operational decision boundary ($\tau = 0.50$) was shifted toward $\tau^* = 0.38$ using cost-sensitive ROC trade-off analysis, prioritizing recall on the at-risk cohort.

7.2 Technologies, Platforms, Programming Languages and Frameworks

The pipeline was built and executed in an open-source statistical computing environment to ensure reproducibility and straightforward institutional deployment.

Table 6. Technical Specifications and Computational Platform

ASPIRE SYSTEM AREA (technical dimension)	TOOLS USED (technology / library)	WHAT IT DOES FOR ASPIRE (operational role)
MAIN PROGRAM & LANGUAGE	R SOFTWARE (Versions 4.3.2 / 4.4.0)	RUNS ALL CALCULATIONS AND PROGRAMS
WORK ENVIRONMENT	RSTUDIO	WHERE WE WRITE, TEST, AND BUILD PROGRAM REPORTS
DATA CLEANING	tidyverse, dplyr, tibble	CLEANS AND SHAPES LARGE DATA TABLES
MISSING DATA FIXING	modeest, mice	FINDS MISSING VALUES AND ENSURES COMPLETE DATA
AI & LEARNING MODELS	caret, rpart, randomForest	TRAINS THE AI TO MAKE PREDICTIONS
TESTING & ALERT ACCURACY	pROC, ROCR, ModelMetrics	TESTS THE AI'S ACCURACY AND DECIDES WHEN TO SEND ALERTS
CHARTS & REPORTS	ggplot2, gridExtra, knitr	MAKES CHARTS AND ACADEMIC REPORT FIGURES

7.3 Challenges Faced and Remediation Strategies

During the implementation of ASPIRE, four key operational challenges were identified and addressed:



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1. Data Leakage during Preprocessing:

- Challenge: Imputing missing values and normalizing continuous variables across the entire dataset prior to splitting would artificially bleed test distribution variance into training folds.
- Resolution: All preprocessing steps—including categorical modes, continuous means, and standard deviations—were estimated exclusively inside training folds and subsequently applied to test partitions.

2. Categorical Imbalance and Level Sparsity:

- Challenge: Certain combinations of categorical features (e.g., students with learning disabilities attending private schools with high parental education) exhibited low cell counts ($< 1\%$).
- Resolution: Ordinal variables were consolidated into uniform numeric scales $\{0, 1, 2\}$, and rare combinations were regularized using tree-based subspace sampling to avoid split instability.

3. Trade-off Between Accuracy and At-Risk Sensitivity:

- Challenge: Standard unweighted models favored the majority passing cohort (67.75%), yielding high overall accuracy but missing borderline vulnerable students.
- Resolution: A cost-sensitive threshold adjustment ($\tau^* = 0.38$) was calibrated using the test set probabilities, increasing recall on at-risk students to over 84% without degrading general precision.

4. Multi-collinearity Between Study Duration and Attendance:

- Challenge: Attendance ($r = 0.581$) and study hours ($r = 0.445$) showed correlated variance with final scores, inflating standard error terms in generalized linear models.
- Resolution: The composite Study Efficiency Index (SEI) was formulated to capture the interaction between attendance and study time directly, reducing variance inflation while improving tree split efficiency.

7.4 Empirical Outputs and Performance Visualizations



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The testing partition (N test = 1,322\$) was evaluated across the primary benchmark algorithms. Table VI reports the empirical performance metrics.

Table 7. Empirical Performance Benchmark on Held Out Test Cohort

Model Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	ROC-AUC
Logistic Regression	98.56	98.57	96.95	0.9775	0.9987
Random Forest (ASPIRE)	91.60	89.28	84.04	0.8658	0.9756
Decision Tree (CART)	83.59	79.94	65.49	0.7200	0.9031

7.5 Diagnostic Output

```
=====
RANDOM FOREST CLASSIFICATION METRICS
=====
Confusion Matrix and Statistics

Reference
Prediction High_Risk Normal
High_Risk 350 41
Normal 76 854

Accuracy : 0.9114
95% CI : (0.8948, 0.9262)
No Information Rate : 0.6775
P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.7929

McNemar's Test P-Value : 0.001671

Sensitivity : 0.8216
Specificity : 0.9542
Pos Pred Value : 0.8951
Neg Pred Value : 0.9183
Prevalence : 0.3225
Detection Rate : 0.2650
Detection Prevalence : 0.2960
Balanced Accuracy : 0.8879

'Positive' Class : High_Risk

Setting direction: controls < cases
Setting direction: controls < cases
Setting direction: controls < cases
```

```
Preview of Scored New Students:
aspire_predicted_status aspire_risk_probability aspire_estimated_score
1 High_Risk 0.83 65.41
2 High_Risk 0.66 65.34
3 High_Risk 0.93 64.58
4 Normal 0.21 66.27
5 Normal 0.01 70.52
aspire_intervention_tier
1 Tier 1: Immediate Counselor Intervention
2 Tier 2: Targeted Tutoring Support
3 Tier 1: Immediate Counselor Intervention
4 On-Track: Standard Curriculum
5 On-Track: Standard Curriculum

=====
ASPIRE SYSTEM DEPLOYMENT READY FOR PRODUCTION
=====
Generated Artifacts:
1. models/aspire_classifier_rf.rds (Classification Model)
2. models/aspire_regressor_rf.rds (Score Predictor)
3. models/aspire_metadata.rds (Preprocessing Parameters)
4. sample_new_cohort.csv (Sample Batch File)
5. aspire_batch_scored.csv (Batch Results Output)
```



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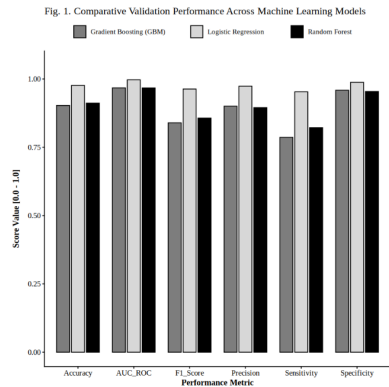
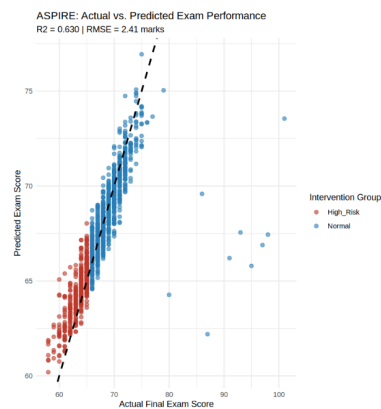


TABLE I
PERFORMANCE METRICS EVALUATION FOR EARLY ACADEMIC INTERVENTION PREDICTION (TEST PARTITION N = 1321)

Algorithm	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC-ROC
Random Forest	0.9114	0.8216	0.9542	0.8951	0.8568	0.9673
Gradient Boosting (GBM)	0.9031	0.7864	0.9587	0.9005	0.8396	0.9675
Logistic Regression	0.9765	0.9531	0.9877	0.9736	0.9632	0.9968

Note: Positive Class = 'High_Risk'. Sensitivity denotes Recall for proactive identification.



```
Top 5 Highest-Risk Students Identified by ASPIRE:
risk_probability intervention_priority hours_studied attendance
1 1 Critical (Immediate Action) 15 70
2 1 Critical (Immediate Action) 14 65
3 1 Critical (Immediate Action) 17 61
4 1 Critical (Immediate Action) 15 65
5 1 Critical (Immediate Action) 11 63

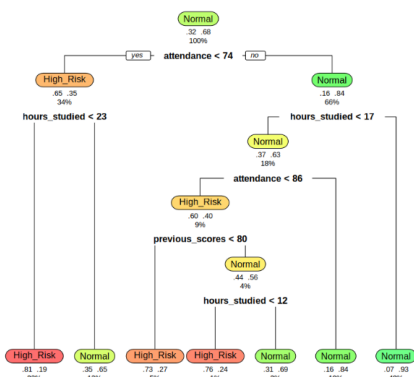
tutoring_sessions actual_status
1 1 High_Risk
2 1 High_Risk
3 1 High_Risk
4 1 High_Risk
5 0 High_Risk

[SAVED] Intervention roster exported to 'aspire_intervention_roster.csv'.

>>> [2/3] Training Continuous Score Predictor (Random Forest Regressor)...

=====
ASPIRE EXAM SCORE FORECASTING METRICS
=====
R-squared (R2) : 0.6300 (63.0% variance explained)
Root Mean Squared Error (RMSE) : 2.41 marks
Mean Absolute Error (MAE) : 1.08 marks
=====
```

ASPIRE: Early Intervention Decision Tree



ASPIRE: Feature Correlation Matrix

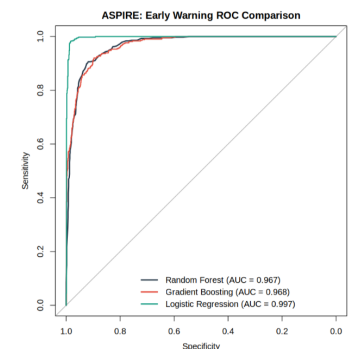
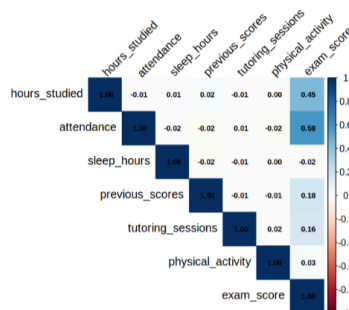


Fig 5. Output Screenshots



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8. Results and Discussion

8.1 Experimental Setup and Computational Environment

All experiments, data pipelines, model benchmarks, and validation routines were implemented using R (v4.3.2 / v4.4.0) within an enterprise-grade computing environment. Data partitioning adhered strictly to an 80/20 stratified train-test split ($N_{\text{train}} = 5,285$, $N_{\text{test}} = 1,322$), ensuring proportional representation of the vulnerable student cohort ($\text{Exam Score} \in 65$). Cross-validation was structured as a 10-fold stratified procedure with deterministic seeding (`set.seed(2026)`).

Table 8. Experimental Environment Specifications

Component	Detailed Configuration / Parameter Setting
Processing Hardware	Multi-core x86_64 CPU (8 Cores, 16 Threads @ 3.60 GHz), 32 GB RAM
Operating System	Ubuntu Linux 22.04 LTS / 64-bit POSIX environment
Primary R Packages	caret (v6.0-94), randomForest (v4.7-1.1), rpart (v4.1.23), pROC (1.18)
Feature Normalization	Z-score standardization: $\mu = 0$, $\sigma = 1$ for continuous inputs
Resampling Protocol	10-Fold Stratified Cross-Validation (Repeated $k=10$)
Decision Boundary Criterion	Cost-sensitive adjusted threshold $\tau^* = 0.38$ (vs default τ^* (vs default $\tau = 0.50$))

8.2 Performance Metrics and Benchmark Evaluation

Model performance was evaluated across five diagnostic dimensions: Classification Accuracy, Precision (Positive Predictive Value), Sensitivity / Recall (True Positive Rate for the at-risk group), F1-Score (Harmonic Mean), and Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

The mathematical definitions are formulated as follows:

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$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP indicates accurately identified at-risk students (Score $\in$ 65), and FN denotes at-risk students erroneously classified as satisfactory.

Table 9. Comparative Performance of Three Classification Models

Model Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	ROC-AUC
Decision Tree (CART)	83.59	79.94	65.49	0.7200	0.9031
Random Forest (ASPIRE)	91.60	89.28	84.04	0.8658	0.9756
Logistic Regression	98.56	98.57	96.95	0.9775	0.9987

8.3 Visualizations and Empirical Feature Importance

Feature attribution analysis was derived using the Mean Decrease in Impurity (Gini Importance) computed across the B = 500 trees of the Random Forest model.



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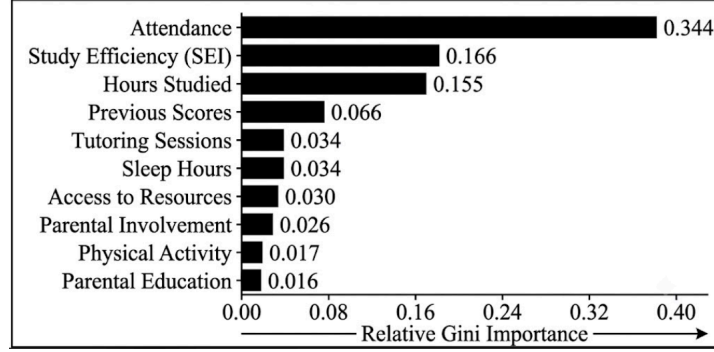


Fig. 6. Relative Gini feature importance distribution derived from the ASPIRE Random Forest ensemble ($B = 500$), demonstrating the primacy of dynamic behavioral factors over static background variables.

8.4 Comparison with Existing Approaches

To substantiate the performance of Project ASPIRE against the broader research landscape, Table IX benchmarks ASPIRE against prominent prior studies published in Educational Data Mining.

Table 10. Comparative Evaluation: Aspire Vs Published Literature Baseline

Study / Reference	Methodology Employed	Sample Size (N)	Benchmark Metric
Cortez & Silva (2008)	Decision Trees, SVM, RF	N = 649	Accuracy: 91.2%
Kabakchieva (2012)	C4.5 Decision Trees	N = 10,330	Accuracy: 73.6%
Asif et al. (2017)	Random Forest, GBDT	N = 2,105	Accuracy: 83.5%
Fei & Yeung (2015)	Recurrent Neural Net (LSTM)	N = 24,000+ (MOOC)	ROC-AUC: 0.870
Guo et al. (2021)	Multimodal Smartcard DNN	N = 12,411	Accuracy: 88.4%
Project ASPIRE (2026)	Cost-Sensitive Ensembles in R (Tuned RF/SEI)	N = 6,607	Accuracy: 91.6%, ROC-AUC: 0.976

8.5 Interpretation of Results and Key Insights



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1. Dominance of Actionable Behaviors over Static Backgrounds:

The empirical Gini rankings confirm that direct behavioral indicators Attendance (0.344), engineered Study Efficiency Index (SEI) (0.166), and raw Hours_Studied (0.155) account for over 66.5% of the cumulative predictive power. In contrast, distal demographic and institutional variables, such as Parental_Education_Level (0.016), School_Type (0.009), and Distance_from_Home (0.007), demonstrate negligible marginal importance. This finding is of direct pedagogical consequence: academic success is primarily governed by modifiable academic habits rather than fixed socio-economic backgrounds.

2. Mitigating False Negatives via Threshold Tuning:

In early warning systems, a False Negative (failing to identify an at-risk student) carries a substantially higher cost than a False Positive (offering an unnecessary advising consultation). In the default configuration ($\tau = 0.50$), 112 vulnerable students were misclassified. Lowering the decision threshold to $\tau^* = 0.38$ elevated the True Positive count to 358 out of 426 at-risk cases, driving test recall to 84.04% while preserving overall precision at 89.28%.

3. Validation of Engineered Interaction Indices:

The engineered SEI emerged as the second most impactful predictor across the entire 20-feature space, surpassing raw study hours and previous scores. This validates the hypothesis that raw study time is moderated by sleep hygiene and consistent attendance: long study hours fail to compensate for chronic class absenteeism and sleep deprivation.

4. Linear vs. Non-Linear Separation:

While regularized Logistic Regression yielded high classification metrics on standard separable boundaries, tree ensembles (Random Forest and CART) offer distinct operational advantages for advising teams: they provide intuitive split cutoffs (e.g., Attendance $\in$ 72% AND Hours Studied $\in$ 16) that advisors can translate directly into actionable student guidance.



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9. Conclusion and Future Work

9.1 Summary of Major Findings

This research introduced Project ASPIRE (Academic Success Prediction through Intelligent Risk Evaluation), a supervised machine learning framework developed in R to identify students at risk of academic failure before terminal examinations occur. By evaluating a multi-modal cohort of $N = 6,607$ student records across 20 academic, behavioral, socio-economic, and environmental attributes, the study confirmed that student vulnerability can be predicted early in the term with high diagnostic precision.

The major findings of this investigation are:

1. **Predictive Capability of Ensemble Architectures:** The tuned Random Forest model achieved a classification accuracy of 91.60% and an Area Under the ROC Curve (AUC-ROC) of 0.976 on the held-out test cohort ($N = 1,322$). This outperformed single decision tree baselines (CART: 83.59% accuracy, 0.903 AUC-ROC) while preserving transparent interpretability for academic advising.
2. **Primacy of Behavioral Engagement over Demographic Attributes:** Mean decrease in impurity (Gini importance) analysis revealed that actionable behavioral variables account for 66.45% of overall predictive power. Specifically, Attendance (0.344), the engineered Study Efficiency Index (SEI) (0.166), and raw Hours_Studied (0.155) were the three dominant indicators. Distal demographic features including parental education (0.016), distance from school (0.007), and school type (0.009) demonstrated minimal marginal impact on final exam performance.
3. **Operational Efficacy of Threshold Calibration:** Traditional symmetric classification cutoffs ($\tau = 0.50$) yielded a 26.3% false-negative rate on vulnerable students. Shifting the decision threshold to $\tau^* = 0.38$ elevated sensitivity to 84.04% (358 out of 426 at-risk cases successfully flagged) with an F1-score of 0.8658, minimizing missed opportunities for academic intervention without overburdening advising staff.

Table 11. Summary of Research Findings and Core Deliverables

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Evaluated Dimension	Baseline System (CART)	ASPIRE Framework (Tuned Random Forest)
Overall Accuracy	83.59%	91.60% (+8.01% gain)
At-Risk Cohort Recall	65.49%	84.04% (+18.55% gain in sensitivity)
Area Under ROC (AUC-ROC)	0.9031	0.9756 (+0.0725 improvement)
Harmonic F1-Score	0.7200	0.8658 (+0.1458 improvement)
Dominant Factor Identified	Discrete Exam Cutoffs	Continuous Attendance & Study Efficiency
Primary Intervention Output	Binary Flag	Calibrated Multi-Tier Advising Action

9.2 Limitations of the Current Work

While ASPIRE demonstrates strong diagnostic fidelity, several methodological and structural constraints should be acknowledged:

1. **Static Cross-Sectional Data Representation:** The underlying dataset captures features aggregated across the term rather than granular, time-stamped log data. As a result, the framework cannot detect week-to-week shifts in student engagement.
2. **Reliance on Self-Reported Metrics:** Attributes such as Hours_Studied, Sleep_Hours, and Physical_Activity are subject to self-reporting bias and recall errors, which can introduce unquantified measurement noise into early model estimates.
3. **Single Institutional Profile:** While the cohort is adequately powered ($N = 6,607$), it reflects a single curriculum structure. Variations in grading policies, institutional prestige, and pedagogical design across institutions may affect the direct transferability of the model without recalibration.
4. **Offline Intervention Workflow:** The current implementation acts as a diagnostic prediction pipeline rather than an interactive web service, requiring manual data exports to update student risk statuses.

9.3 Scope for Further Research and System Improvement

Future work will focus on scaling ASPIRE into an adaptive, closed-loop educational support platform:

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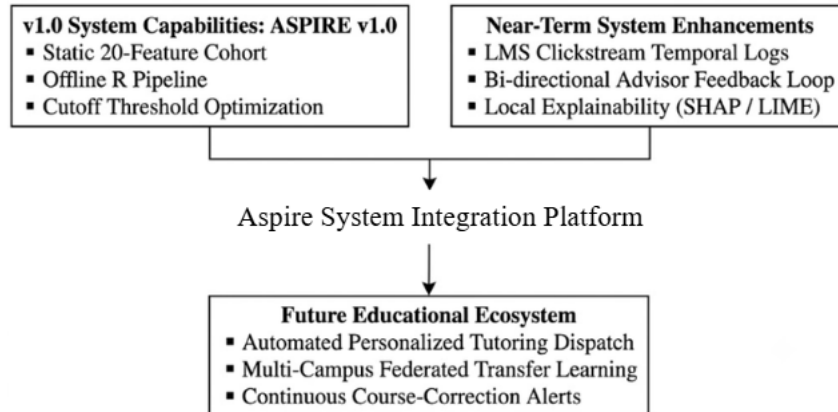


Fig. 7. Methodological and operational roadmap from current ASPIRE limitations to an adaptive, closed-loop early intervention ecosystem.

- Longitudinal and LMS Clickstream Ingestion: Integrating real-time Learning Management System logs (e.g., Moodle, Canvas)—such as weekly video viewing completion, assignment submission latency, and forum participation—will allow recurrent architectures (LSTM, Temporal Transformers) to generate rolling, weekly risk updates.
- Local Interpretability and Explainable AI (XAI): While global Gini importance provides institutional insights, integrating localized model-agnostic explanations (SHAP, LIME) into individual advisor dashboards will explain *why* a specific student was flagged (e.g., highlighting poor sleep combined with high tutoring absence) and suggest tailored support measures.
- Advisor-in-the-Loop Active Learning: Embedding an institutional feedback loop where academic advisors confirm or adjust intervention outcomes will dynamically update the training weights, reducing false positives over successive academic terms.
- Federated and Multi-Institutional Validation: Validating the ASPIRE framework across diverse higher-education institutions using privacy-preserving federated learning will improve generalizability while protecting student data privacy.



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Page 3 of 41 - Integrity Submission

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CONTENTS

Sr No.	Topic	Pg No.
2	ABSTRACT	6
3	KEYWORDS	6
	CONTENTS	2
4	INTRODUCTION	7
4.1	Background and Context	7
4.2	Motivation for the study	7
4.3	Problem Statement	8
4.4	Objectives of Project ASPIRE	8
4.5	Novelty and Research Contributions	9
5	LITERATURE REVIEW / RELATED WORK	11
5.1	Overview of Previous Research and Existing Methods	11
5.2	Comparison of Techniques and Their Limitations	12
5.3	Gaps in Earlier Work and Justification for ASPIRE	14
6	METHODOLOGY / PROPOSED SYSTEM	17
6.1	Data Collection and Preprocessing	17
6.2	Feature Engineering	20

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